

```
In [3]: import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
In [4]: df = pd.read_csv(r"C:\Users\LIKHITH\Downloads\titanic_ dataset_final.csv")
```

```
In [18]: df
```

Out[18]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05
...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.73

891 rows × 12 columns



```
In [20]: # Display dataset info
print(df.describe())
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
In [22]: # Missing Values Check
print("\nMissing Values:\n", df.isnull().sum())
```

```
Missing Values:
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

```
In [28]: # Survival Rate Visualization
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
```

```
Out[28]: Text(0.5, 1.0, 'Survival Count')
```

```
In [29]: plt.show()
```

```
<Figure size 640x480 with 1 Axes>
```

```
In [12]: import ollama
def generate_insights(df_summary):
    prompt = f"Analyze the dataset summary and provide insights:\n\n{df_summary}"
    response = ollama.chat(model = "gemma3:270m", messages = [{"role": "user", "content": prompt}])
    return response ['message'] ['content']
```

```
summary = df.describe().to_string()
insights = generate_insights(summary)
print("\n AI-generated Insights:\n", insights)
```

AI-generated Insights:

Okay, I've analyzed the dataset summary. Here's a breakdown of the key insights:

```
*  **Overall Trend:** The dataset shows a positive trend, with a relatively consist
ent average value for the `Age` and `SibSp` values. This suggests a stable and well-
functioning population.
*  **Age Distribution:** The `Age` values are generally increasing, suggesting a gr
owing population. The `SibSp` values are relatively stable, indicating a stable popu
lation structure.
*  **Per Capita Income:** The `Per Capita Income` value is consistently above the `
Mean` value, indicating a relatively high level of population income.
*  **Population Density:** The `Population Density` value is generally above the `S
td Deviation` value, suggesting a relatively low population density.
*  **Health Status:** The `Health Status` value is generally above the `Mean` valu
e, indicating a relatively healthy population.
*  **Social Segregation:** The `Social Segregation` value is generally above the `M
ean` value, indicating a relatively low social segregation.
```

In summary, the dataset showcases a healthy and growing population with a relatively stable average age and a generally healthy population with a relatively low social segregation. The `Per Capita Income` value indicates a relatively high level of population income.

```
import gradio as gr
def eda_analysis(file):
    df = pd.read_csv(file.name)
    summary = df.describe().to_string()
    insights = generate_insights(summary)
    return insights
demo = gr.Interface(fn=eda_analysis, input="file", output="text", title="AI-powered Eda With gemma by likhith")
demo.launch(share=True)
```

```
In [17]: import gradio as gr
import pandas as pd

# Dummy insight generator (replace with Gemma / LLM Logic later)
def generate_insights(summary):
    return f"EDA Summary Insights:\n\n{summary}"

def eda_analysis(file):
    df = pd.read_csv(file.name)
    summary = df.describe().to_string()
    insights = generate_insights(summary)
    return insights

demo = gr.Interface(
    fn=eda_analysis,
    inputs=gr.File(file_types=[".csv"]),
    outputs="text",
    title="AI-powered EDA with Gemma by Likhith"
)

demo.launch(share=True)
```

* Running on local URL: <http://127.0.0.1:7860>
* Running on public URL: <https://8e186bc2dfdd4eff56.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run ``gradio deploy`` from the terminal in the working directory to deploy to Hugging Face Spaces (<https://huggingface.co/spaces>)



No interface is running right now

Out[17]:

In []: